Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming I part–July 24th, 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Explain the factors that simplify the computation of time complexity in a PRAM application, giving them in order of priority. (1)

B. Please define what "Cost" means in a PRAM application. (1)

C. Please describe how **matrix x matrix** works when considering a PRAM architecture. (1)

D. Please describe what happens in a PRAM model when the number of threads exceeds the number of processors. (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the cases the master theorem solves and under which assumption the theorem holds. Apply then to the powering algorithm. Is the master theorem of any help if we would like to estimate the spatial complexity of an algorithm? (2)

B. Please describe the "Select the i th smallest of n elements" algorithm and its key properties. How is it randomized? (1) **This question can be skipped in case of a passed challenge.**

C. To which class does the Move-To-Front Algorithm belong? How is its complexity evaluated? (1)

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss how a Union is implemented in a Treap Data-Structure. How does the number of rotations impact the search for an element in a Treap? (2)

B. Describe how the insertion for skip-list data structure works. What about its complexity? (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the key property of the longest common subsequence. In which class of algorithm can be inserted? Why? (2)

B. What is meant by "amortized analysis"? To which type of algorithms could be applied? Examples? (2)